A CASE-STUDY BASED REPORT

ON

Traffic Flow Data Visualization Using PCA and t-SNE

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Abstract

Modern traffic management systems collect data describing vehicle movement, congestion, road occupancy, communication quality and environmental conditions. Such data is high-dimensional, so examining relationships among observations directly is difficult. This project, “Traffic Flow Data Visualization Using PCA and t-SNE,” studies how dimensionality-reduction methods can make the structure of traffic-flow data easier to inspect. The dataset contains 195,714 observations, 27 columns, 24 numerical features and records from 500 road segments. Exploratory Data Analysis (EDA) was used to inspect the dataset, feature types, traffic-condition labels and data quality. Numerical features were standardized before dimensionality reduction. Principal Component Analysis (PCA) was applied to the complete dataset, and its first two components explained 92.53% of the variance. Because t-SNE is computationally expensive on a dataset of this size, a stratified sample of 5,000 observations was used to create a nonlinear two-dimensional embedding. The t-SNE plot showed Free-flow observations mostly separated from other conditions, while Heavy and Gridlock observations overlapped. PCA and t-SNE therefore provide complementary views of high-dimensional traffic data. This is an exploratory visualization project, not a traffic prediction or classification system.

1. Introduction

1.1 Background

Traffic monitoring systems can record many measurements at the same time, including average vehicle speed, vehicle density, waiting time, occupancy, flow, queue length, vehicle movement, communication delay, packet loss and weather-related information. Each record can be represented by a vector of numerical features. When many features are considered together, it is difficult to understand relationships between observations using ordinary tables or two-dimensional plots.

Dimensionality reduction transforms data with many features into a smaller representation. PCA is a linear technique that constructs components capturing variation in the data. t-SNE is a nonlinear technique designed to represent local similarities between observations in a lower-dimensional space. Visualizing the resulting two-dimensional representations helps analysts explore patterns and overlap among traffic conditions.

1.2 Motivation for the Project

The motivation is to make a large traffic dataset easier to interpret visually. A traffic analyst may want to explore whether observations associated with Free-flow, Moderate, Heavy and Gridlock conditions occupy different regions of a feature space, and whether some conditions have similar characteristics. Direct plotting of 24 numerical dimensions is impractical, so PCA and t-SNE are used to produce two-dimensional views.

1.3 Overview of the Work Carried Out

- Loaded the traffic-flow dataset in Google Colab and inspected its dimensions and columns.
- Reviewed feature types, traffic-condition label counts, missing values and duplicate records.
- Separated the 24 numerical input features from timestamp, road-segment identifier and label columns.

– Standardized numerical features so different measurement scales would not dominate the analysis.
– Applied PCA to the complete dataset and reviewed the explained-variance ratio of its first two components.
– Selected a stratified sample of 5,000 observations for t-SNE and visualized the two-dimensional embedding.
– Compared the methods and interpreted the resulting plots.

1.4 Scope and Responsible-Use Statement

The scope is limited to exploratory analysis and two-dimensional visualization of the supplied dataset. The project does not train a supervised prediction model, estimate future congestion, recommend signal timings or control real traffic infrastructure. Labels are used to colour and interpret plots; they are not included as input features to PCA or t-SNE. Visual separation in a two-dimensional embedding is not proof that classes are perfectly separable in the original feature space. Operational use would require additional validation, domain review and testing on suitable real-world data.

2. Problem Statement

2.1 Case-Study Scenario

Consider a smart-city traffic management team that receives observations from hundreds of road segments. Each observation contains measurements related to traffic flow, vehicle behaviour, network communication and environmental conditions. The dataset has 24 numerical features per observation and nearly two hundred thousand observations. The team wants to understand the overall structure of the data and visually examine how observations labelled Free-flow, Moderate, Heavy and Gridlock relate to one another. Viewing each feature separately does not provide a compact picture of the multivariate relationships.

2.2 Central Case-Study Question

How can high-dimensional traffic-flow data be reduced to a lower-dimensional representation while retaining important information, so that different traffic conditions and their patterns can be visualized and analyzed effectively?

2.3 Formal Problem Definition

Given a dataset X containing 195,714 observations and 24 numerical features, the task is to transform feature vectors into a two-dimensional representation suitable for visualization. The analysis should retain important structure, make broad variation visible, and allow traffic-condition labels to be used for interpretation. PCA represents overall variance linearly, while t-SNE explores local relationships in a nonlinear embedding of a representative sample.

2.4 Objectives that Guide the Study

- Inspect the dataset and summarize its size, features, labels and data quality.
- Prepare numerical inputs for dimensionality-reduction algorithms.
- Use PCA to reduce the 24-dimensional data to two components and measure variance explained.
- Use t-SNE on a stratified sample to explore nonlinear local relationships in two dimensions.

- Compare the methods in terms of purpose, output and interpretation.
- Document observations and limitations without claiming predictive performance.

2.5 Assumptions and Limitations

- The supplied dataset is treated as the source of observations and labels for this study.
- The 24 selected numerical columns are features; timestamp, road_segment_id and label are excluded from the numerical feature matrix.
- Standardization is applied because the numerical columns use different scales.
- PCA is applied to the full dataset, whereas t-SNE uses a 5,000-observation stratified sample due to computational cost.
- t-SNE axes do not have a directly comparable explained-variance interpretation to PCA axes.
- The plots are exploratory; overlap or separation in two dimensions is not a formal classification score.
- Results depend on the dataset, preprocessing and algorithm settings.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The project uses vanet_traffic_data.csv, which contains traffic-related observations from a vehicular network context. The dataset combines traffic measurements, vehicle movement indicators, communication/network measurements and environmental variables.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

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Dataset property	Observed value
Number of observations	195,714
Number of columns	27
Numerical features selected	24
Unique road segments	500
Unique timestamps	195,714
Traffic-condition labels	4
Missing values	0
Duplicate records	0

The four traffic-condition labels and their observed counts are shown below.

Traffic condition	Number of observations

Free-flow	57,812
Moderate	54,018
Heavy	47,440
Gridlock	36,444
Total	195,714

3.2 Purpose of EDA

EDA was performed before dimensionality reduction to understand the shape and quality of the data, identify the role of each column, examine the distribution of traffic-condition labels and check for missing or repeated records. It also helped establish which columns should be used as numerical features and which should be retained only for identification or interpretation.

3.3 Preparation Steps Actually Performed

- Loaded the CSV dataset into a pandas DataFrame in Google Colab.
- Inspected the dataset dimensions and column names.
- Reviewed data types and identified timestamp, road_segment_id and label as contextual columns.
- Checked missing values and duplicate rows; both reported counts were zero.
- Reviewed the distribution of the four traffic-condition labels.
- Created the numerical feature matrix by excluding timestamp, road_segment_id and label.
- Standardized numerical features before PCA and t-SNE.

3.4 EDA Exploration – Implementation

The numerical feature matrix was defined by excluding the three contextual columns from the DataFrame. The label column was kept separately so that it could be used to colour visualizations after the embeddings were calculated.

```
X = df.drop(columns=["timestamp", "road_segment_id", "label"])\ny = df["label"]
```

The EDA checks established that the dataset has 195,714 rows and 27 columns, including 24 numerical features. The label counts were reviewed to understand representation of each condition. The reported missing-value and duplicate-row checks both returned zero.

3.5 Summary of EDA Findings

- The dataset contains 24 numerical features describing different aspects of traffic and its operating environment.
- All four traffic labels are represented; Free-flow has the largest count and Gridlock the smallest.
- No missing values or duplicate rows were reported in the checks performed.
- Timestamp and road-segment identifier are contextual fields, not numerical features for this analysis.
- The label is used to interpret plots but is excluded from the feature matrix, keeping dimensionality reduction unsupervised.

4. Algorithm Selection and Implementation

4.1 Workflow and Preprocessing

The workflow is: load dataset → inspect and check the data → separate numerical features and labels → standardize numerical features → apply PCA to the full dataset → apply t-SNE to a stratified sample → plot the two-dimensional embeddings → compare and interpret the results. The dimensionality-reduction stage is unsupervised because the algorithms transform feature data without using labels to learn class boundaries.

Standardization places numerical features on a comparable scale. This matters because columns such as speed, temperature, communication delay and packet-loss percentage can have different units and ranges. Without standardization, larger numerical scales could disproportionately influence the transformation.

4.2 Principal Component Analysis (PCA)

PCA is a linear dimensionality-reduction technique. It constructs new axes called principal components as combinations of the original features. The first component captures the largest possible amount of variance; each subsequent component captures as much remaining variance as possible while being orthogonal to earlier components. In this project, the 24 standardized numerical features were transformed and the first two components were used as coordinates for a two-dimensional scatter plot.

The explained-variance results from the full-dataset PCA run were:

Component	Variance explained
PC1	83.68%
PC2	8.85%
PC1 + PC2	92.53%

Interpretation: the first two components account for 92.53% of the total variance in the standardized dataset. This means the two-dimensional PCA view retains a large proportion of overall linear variation. It does not mean that 92.53% of traffic labels are classified correctly; explained variance is not classification accuracy.

4.3 t-distributed Stochastic Neighbor Embedding (t-SNE)

t-SNE is a nonlinear dimensionality-reduction method commonly used to visualize high-dimensional data. It aims to place observations with similar local neighbourhoods near one another in a lower-dimensional space. Unlike PCA, t-SNE does not provide an explained-variance ratio for its two axes, and distances between far-apart clusters should be interpreted cautiously.

Because the complete dataset contains 195,714 observations, t-SNE was run on a representative stratified sample of 5,000 observations. The sample composition was:

Traffic condition	Sample observations
Free-flow	1,477
Moderate	1,380
Heavy	1,212
Gridlock	931

Total	5,000

The sampled numerical features were standardized and embedded into two dimensions, represented by tSNE1 and tSNE2. The scatter plot was coloured using the existing labels so the arrangement of observations for each condition could be inspected.

4.4 Visualization and Interpretation

The PCA plot provides a compact view of broad variation in the full feature dataset. Its axes correspond to principal components, and the variance-explained values quantify how much overall variation is represented by the first two axes.

In the t-SNE visualization, Free-flow observations are mostly located separately from the other traffic conditions, while Heavy and Gridlock observations show overlap. This suggests that some Free-flow observations have distinguishable local relationships in the selected feature space, while Heavy and Gridlock observations share characteristics that make their embedded points less distinct. The visualization does not establish perfect separability or demonstrate that the system can predict traffic labels.

4.5 PCA and t-SNE Comparison

Aspect	PCA	t-SNE
Type	Linear dimensionality reduction	Nonlinear dimensionality reduction
Main emphasis	Overall variance	Local neighbourhood relationships
Input used	Complete dataset	Stratified sample of 5,000 observations
Output	Principal-component coordinates	Two-dimensional embedding
Quantitative summary	PC1 and PC2 explain 92.53% combined variance	No directly comparable explained-variance percentage
Use in this project	Compact global view of variation	Explore local patterns and groupings
Caution	Variance explained is not classification accuracy	Visual separation is not proof of class separability

The methods are complementary rather than interchangeable. PCA gives an interpretable summary of global variation, while t-SNE can reveal local groupings that may be difficult to observe in a linear projection. Using both provides a more informative exploratory view than relying on either plot alone.

5. Conclusion

This case study explored how high-dimensional traffic-flow data can be visualized using PCA and t-SNE. The dataset contains 195,714 observations, 24 numerical features and four traffic-condition labels across 500 road segments. EDA checks reported no missing values and no duplicate records. Numerical features were separated from contextual columns and standardized before dimensionality reduction.

PCA was applied to the complete dataset. Its first two components explained 83.68% and 8.85% of the variance respectively, for a combined 92.53%. This supports representing much of the overall variation in a two-dimensional linear projection. t-SNE was applied to a stratified sample

of 5,000 observations because of the computational cost of processing the full dataset. Its visualization showed Free-flow observations mostly separated from other conditions, while Heavy and Gridlock observations overlapped.

The project demonstrates that PCA and t-SNE offer complementary perspectives for exploring traffic data. PCA summarizes broad variation, whereas t-SNE helps inspect local relationships. The work remains an exploratory visualization study and does not claim predictive performance. Future extensions could investigate additional sampling strategies, compare embeddings across settings, or develop a separately evaluated prediction model if the project scope is expanded.

References

1. Project dataset: vanet_traffic_data.csv, used for the traffic-flow visualization study.

2. Project case-study material supplied with the project: “Traffic Flow Data Visualization Using PCA and t-SNE.”

3. Scikit-learn documentation for PCA, StandardScaler and t-SNE may be consulted for implementation details. Numerical results in this report follow the supplied project results.